SAC's Entropy Coefficient as an Implicit Success Signal in Robotic Manipulation

Anonymous Author(s)
Affiliation
Address
email

Abstract: Soft Actor-Critic (SAC) is the standard algorithm for continuous-control robotic manipulation, yet practitioners routinely override its auto-tuned entropy coefficient α with fixed schedules or manual values, assuming auto-tuning is merely adequate. We challenge this assumption through a controlled empirical study of 128 training runs across five Meta-World manipulation tasks, with comprehensive logging of α trajectories, Q-values, and replay buffer composition. We discover that auto-tuned α acts as an *implicit indicator of training progress*: on challenging tasks, α bifurcates into distinct regimes that predict outcome hundreds of thousands of steps before reward differences emerge. Solved seeds maintain α in a stable moderate range (0.02–0.25), while failed seeds exhibit either α-*collapse* ($\alpha \to 0$, premature determinism) or α-*explosion* ($\alpha \to 9+$, entropy dominance). A controlled 2×2 ablation (64 runs) shows that fixed entropy annealing forces all seeds into the collapse regime, reducing final returns by up to 98% on pick-place. Using a two-sided threshold ($0.005 < \alpha < 1.0$), α alone correctly classifies all 24 seeds by whether they *ever* solve the task (100%), with three discover-then-forget seeds whose α stays in the healthy range despite eventual policy reversion representing a separate failure mode that requires joint monitoring of α and reward to detect.

Keywords: reinforcement learning, soft actor-critic, entropy regularization, robotic manipulation, training diagnostics

1 Introduction

Soft Actor-Critic [1] has become the de facto algorithm for continuous-control robotic manipulation, combining sample efficiency with stability through maximum entropy reinforcement learning. When trained with `ent_coef='auto'`, α is learned via dual gradient descent to maintain a target policy entropy [2]. Despite this principled formulation, practitioners frequently override auto-tuning with fixed values, linear annealing schedules [3], or meta-learned temperatures [4, 5]. The implicit assumption is that auto-tuning is a reasonable default but not fundamentally important—a hyperparameter to be tuned rather than a signal to be observed.

We challenge this assumption with a large-scale diagnostic study. Over 128 training runs on five Meta-World [6] manipulation tasks, we find that auto-tuned α does substantially more than stabilize entropy: it encodes the agent's learning trajectory in a form that predicts eventual task success or failure. On challenging tasks such as *pick-place-v3* and *peg-insert-side-v3*, α trajectories bifurcate into distinct attractors by 200k–300k training steps—well before reward-based diagnostics reveal any meaningful difference between seeds. This bifurcation is not a statistical artifact: the final α value alone classifies seeds as having ever-solved or never-solved with 100% accuracy on both hard tasks (24/24 seeds) using a two-sided criterion, with three discover-then-forget seeds—whose α remains healthy despite eventual policy reversion—constituting a separate failure mode requiring joint monitoring of α and reward.

We identify three qualitatively distinct failure modes, all readable from the α trajectory. **α-collapse**: the coefficient decays to near-zero early in training, the policy becomes near-deterministic, and the agent is trapped in a local minimum for the remainder of training. The mechanism is a negative feedback loop: without task success, Q-values stagnate, the policy gradient carries little signal, the policy stops changing, entropy drifts below target, and α is driven to zero—eliminating the exploration pressure needed to escape. **α-explosion**: the coefficient grows unboundedly to values above 9, the entropy term dominates the objective, and the policy cannot commit to any action. **Discover-then-forget**: α remains in the healthy moderate range, yet the policy catastrophically loses a previously-learned solution. This third mode requires joint monitoring of α and reward to detect; α alone cannot identify it.

To establish causality, we conduct a 2×2 ablation comparing auto-entropy tuning against a fixed annealing schedule, crossed with standard reward and a self-bootstrapped demo reward. Fixed annealing (α: $0.1 \to 0.005$ over 500k steps) reduces solve rates and final returns on both hard tasks. On pick-place, this reduction is catastrophic: 0/8 seeds solve with annealing versus 4/8 with auto-tuning, and mean final

reward drops from 1525 to 33—a 98% reduction. The demo reward does not rescue annealing failure: the demo + anneal condition performs as poorly as anneal alone.

Contributions:

1. We document the α-bifurcation phenomenon across five Meta-World tasks (8–12 seeds per task, 128 total runs), showing that it is tied to task difficulty and reward structure.
2. We identify three failure modes—collapse, explosion, and discover-then-forget—readable from α trajectories, and characterize their mechanistic signatures in Q-values and replay buffer composition.
3. We show via controlled ablation that fixed entropy annealing destroys SAC's adaptive mechanism, causing universal α-collapse and up to 98% performance degradation.
4. We demonstrate that final α predicts whether a seed will ever solve the task with 100% accuracy (24/24 seeds) on hard tasks using a two-sided threshold; discover-then-forget seeds (3/24) constitute a separate failure mode requiring joint α–reward monitoring.

2 Related Work

2.1 Entropy Regularization in RL

Maximum entropy reinforcement learning [7] augments the standard reward objective with a policy entropy bonus, encouraging exploration and robustness to reward misspecification. SAC [1] instantiates this framework for continuous control with a fixed temperature parameter; the subsequent SAC v2 [2] introduced automatic temperature tuning via constrained dual optimization, observing that the choice of α is critical and proposing to learn it online. Eysenbach and Levine [8] provided theoretical grounding for maximum entropy RL by showing it provably solves a class of robust RL problems. Our work extends these foundations by demonstrating that the auto-tuned α carries diagnostic information beyond its intended role as a stability mechanism—it is an emergent signal of training progress.

2.2 Entropy Scheduling and Meta-Learning

Several works have proposed alternatives to auto-entropy tuning, implicitly treating it as insufficient. Zhou and Heywood [3] proposed target entropy annealing for discrete SAC, scheduling the entropy target to decrease over training. Wang and Ni [4] replaced auto-tuning with a metagradient-based approach that learns a schedule for the temperature. Zhou and Heywood [5] proposed a neural network to optimize α in a state-dependent manner. All of these works assume that auto-tuning's constant-target-entropy behavior is a limitation to be overcome. Our evidence suggests the opposite: on dense-reward manipulation tasks, auto-tuning's adaptive behavior is precisely what enables task discovery, and replacing it with a fixed schedule universally degrades performance.

2.3 Failure Modes in Deep RL

Several failure modes of deep RL are relevant to our findings. Nikishin et al. [9] identified the *primacy bias*: early training experiences disproportionately influence the value function, causing the agent to overfit before task-relevant experience arrives. α-collapse may serve as an early indicator of this phenomenon, as low entropy prevents the policy from gathering the diverse experience needed to overcome early overfitting. Lyle et al. [10] characterized *plasticity loss*, whereby neural networks lose their capacity to represent new solutions over time; this is consistent with our discover-then-forget failure mode, where a learned solution is subsequently overwritten. Q-value overestimation [11] is also relevant: our Q-probe analysis shows that annealed conditions (low α) produce Q-value spikes up to 12,000–17,000 followed by collapse, consistent with overestimation under near-deterministic policies.

2.4 Benchmarks and Evaluation

We evaluate on Meta-World v3 [6], a widely-used benchmark for single-task manipulation with 50 distinct tasks using a Sawyer robot arm. McLean et al. [12] recently documented undocumented behavioral changes across Meta-World versions; we use v3 throughout for consistency. Rigorous evaluation methodology following Agarwal et al. [13] motivates our use of 8–12 seeds per condition and per-seed analysis over aggregate statistics.

3 Background

3.1 Soft Actor-Critic with Auto-Entropy Tuning

SAC maximizes the maximum-entropy objective:

$$J(\pi) = \sum_t \mathbb{E}_\pi \big[r(s_t, a_t) + \alpha\, \mathcal{H}(\pi(\cdot|s_t)) \big], \qquad (1)$$

where $\mathcal{H}(\pi(\cdot|s_t)) = -\mathbb{E}_{a\sim\pi}[\log \pi(a|s_t)]$ is the policy entropy and α is the entropy coefficient.

Auto-entropy tuning [2] treats α as a Lagrange multiplier enforcing a minimum entropy constraint. The dual objective is:

$$J(\alpha) = \mathbb{E}_{a\sim\pi} \big[-\alpha \big(\log \pi(a|s) + \bar{\mathcal{H}} \big) \big], \qquad (2)$$

where $\bar{\mathcal{H}}$ is the target entropy (default in SB3 [14]: $\bar{\mathcal{H}} = -\dim(\mathcal{A}) = -4$ for Meta-World's 4-dimensional action

space). When policy entropy $\mathcal{H}(\pi)$ falls below $\bar{\mathcal{H}}$, the gradient on $\log \alpha$ is positive, pushing α up and increasing exploration pressure. When entropy exceeds $\bar{\mathcal{H}}$, α decreases to encourage exploitation. This creates a feedback loop between policy behavior and the entropy coefficient—one whose dynamics, we show, encode training progress.

3.2 Meta-World v3

Meta-World [6] provides 50 manipulation tasks with a Sawyer robot arm. State observations are 39-dimensional; actions are 4-dimensional. Episodes last up to 500 steps; we define success as episode reward ≥ 500.

4 Experimental Setup

4.1 Observational Study: α Trajectories (Exp. 36)

To characterize α dynamics across seeds and tasks, we trained SAC with auto-entropy tuning on five Meta-World v3 tasks, using 8 seeds for easy/medium tasks and 12 seeds for the two hardest tasks (peg-insert-side-v3, pick-place-v3), totaling 64 observational runs. We implemented an `AlphaTrajectoryCallback` that logs $\alpha = \exp(\log \alpha)$ every 1,000 training steps. Additional logs include per-episode success flags, replay buffer success fraction every 50k steps, and Q-values at a fixed set of probe states.

Table 1: Meta-World v3 tasks used in this study.

Task	Category	Difficulty	Seeds
peg-insert-side-v3	Precision	Hard	12
pick-place-v3	Grasping	Hard	12
door-open-v3	Articulated	Medium	8
drawer-close-v3	Simple	Easy	8
window-open-v3	Articulated	Medium	8

4.2 Causal Ablation: Fixed vs. Auto Entropy (Exp. 35)

To test whether auto-entropy tuning is load-bearing, we conducted a 2×2 factorial experiment on the two hard tasks. The factors were: (1) **reward**: environment reward only vs. environment + self-bootstrapped demo reward; (2) **entropy**: auto-tuned vs. fixed annealing schedule. This yielded four conditions (8 seeds each, 64 runs total):

- **A**: Standard SAC with auto-entropy (our primary baseline)
- **B**: SAC with fixed entropy annealing (α: 0.1 → 0.005 over 500k steps)
- **C**: SAC with demo reward and auto-entropy
- **D**: SAC with demo reward and fixed annealing

The demo reward uses a self-bootstrapped k-NN replay buffer (k=5, σ=0.30, scale = 0.5). All other hyperparameters are identical across conditions.

4.3 Shared Configuration

Table 2: Hyperparameters shared across all experiments.

Parameter	Value
Algorithm	SAC (Stable-Baselines3 [14])
Policy network	MLP, 2×256 hidden units
Learning rate	3×10^{-4}
Batch size	256
Replay buffer	10^6 transitions
Discount (γ)	0.99
Soft update (τ)	0.005
Target entropy	$\bar{\mathcal{H}} = -\dim(\mathcal{A}) = -4$
Training steps	10^6
Eval frequency	Every 10^4 steps, 20 episodes
Success threshold	Episode reward ≥ 500

5 Results

5.1 The α Bifurcation

Figure 1 shows α trajectories and reward curves for all 12 seeds on peg-insert-side-v3 and pick-place-v3. On both tasks, seeds bifurcate into qualitatively distinct regimes. Solved seeds (warm lines) converge to a stable moderate range—$\alpha \in [0.017, 0.184]$ for peg-insert and $\alpha \in [0.026, 0.184]$ for pick-place—and maintain this range throughout training. Failed seeds diverge to extreme values: either near-zero or greater than 9. This bifurcation is visible by approximately 200k–300k training steps, whereas reward differences between seeds do not clearly emerge until 400k–600k steps.

Critically, easy tasks (drawer-close, window-open, door-open) show no bifurcation: all 8 seeds on each task converge to moderate α values and solve the task. The bifurcation is a property of task difficulty, not of the algorithm—it requires a reward landscape where some seeds fail to discover the solution within the training budget.

5.2 Three Failure Modes

Pick-place-v3 (12 seeds, 6 failures: 3 never-solved + 3 DTF) exhibits all three failure modes we identify. Figure 2 illustrates each:

α-collapse (seeds 3 and 6 on pick-place; seed 4 on peg-insert): α decays monotonically to near-zero (0.0006–0.0015) within 200k steps. The policy becomes near-deterministic; episode reward stays flat near zero for the remaining 800k steps. This is the most common never-

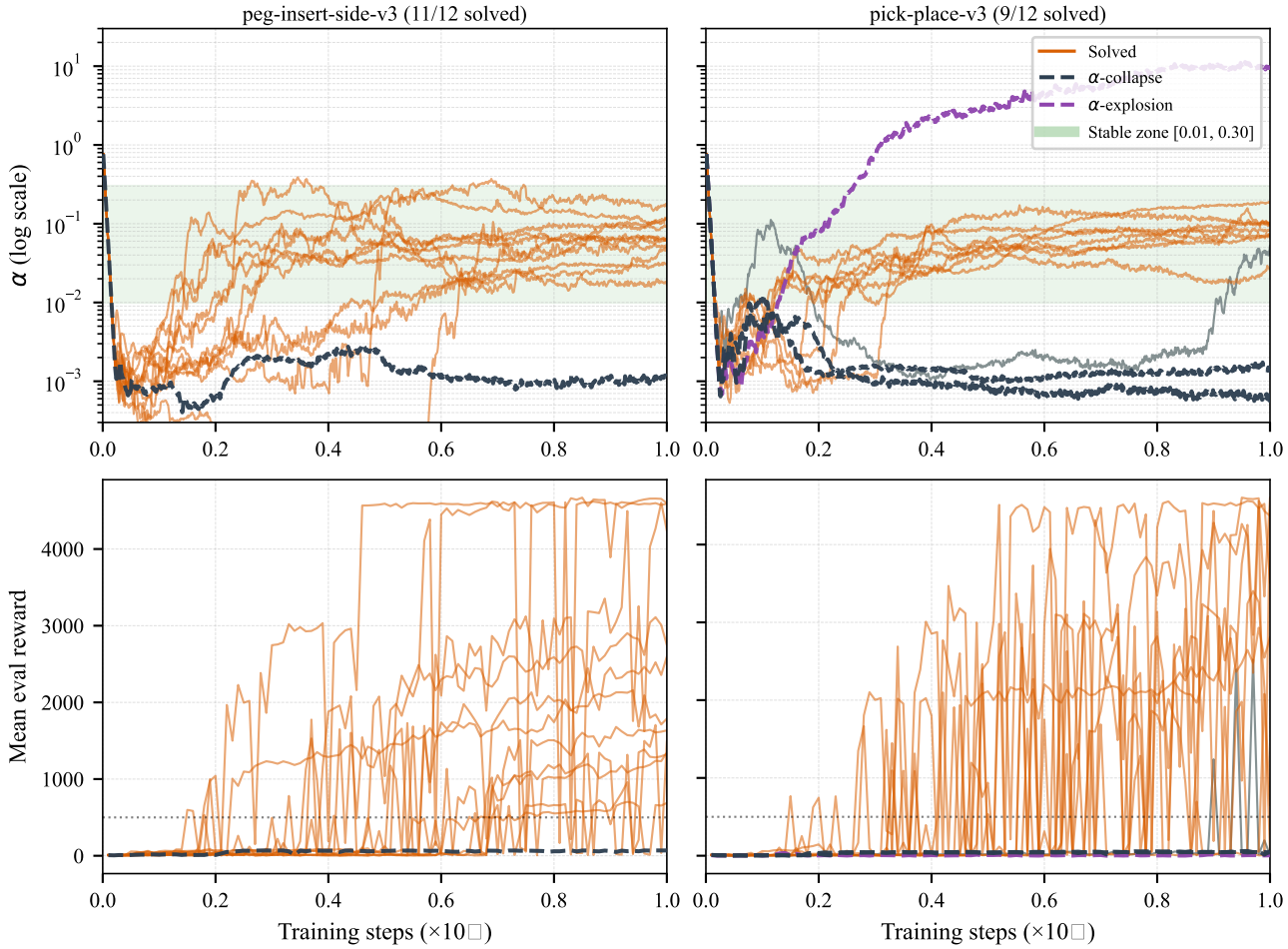

Figure 1: α trajectories (top) and corresponding reward curves (bottom) for peg-insert-side-v3 (left, 12 seeds) and pick-place-v3 (right, 12 seeds). Warm lines indicate ever-solved seeds (including 3 DTF seeds on pick-place); dark blue dashed lines indicate α-collapse; purple dashed lines indicate α-explosion. On both tasks, ever-solved seeds maintain α in the shaded stable zone [0.01, 0.30], while never-solved seeds diverge to extreme values. The α bifurcation is visible by $\sim$200k steps, substantially earlier than reward differences emerge.

solved failure mode (3 of 4 never-solved seeds; counting DTF, there are 6 total failures on pick-place).

α-explosion (seed 1 on pick-place only): α grows from its initial value of $\sim$0.97 to 9.195 by the end of training—an order of magnitude above any solved seed. The entropy term dominates the objective, preventing the policy from committing to effective actions. This rare failure mode (1/24 seeds on hard tasks) is visually striking: while solved seeds' α curves turn down, this seed's α turns sharply up. We hypothesize that this seed's initial random policy happened to exhibit persistently high entropy—above the target $\bar{\mathcal{H}} = -4$—causing the α optimizer to push α downward initially, but as Q-values stagnate without task discovery, the policy gradient stops driving entropy below target, and the auto-tuning enters a runaway positive loop. This represents the dual instability to collapse: collapse occurs when entropy is chronically below target; explosion when it remains chronically above.

Discover-then-forget (seeds 4, 10, and 11 on pick-place): α remains in the moderate range for all three seeds (final values 0.053, 0.026, 0.067), yet each policy discovers a near-solution and then catastrophically reverts: seed 4 peaks at reward ≈ 2519 (step 891k) before reverting to near-zero; seed 10 peaks at ≈ 1618 (step 320k), final reward 16; seed 11 peaks at ≈ 2080 (step 990k), final reward 49. Since α reflects exploration health rather than solution retention, it cannot distinguish these seeds from stably-solved ones—joint monitoring of α and reward is required.

Table 3: Per-task summary across all experimental runs. α statistics are computed from exp. 36 (observational study). Reward statistics are mean $\pm$ std across seeds at final checkpoint. † denotes 12-seed runs. Solve rate counts seeds that ever achieve episode reward ≥ 500 during training; three pick-place seeds (4, 10, 11) discover then forget the solution (DTF).

Task	Solve rate	Final reward mean $\pm$ std	Final α Solved	Final α Failed
peg-insert-side-v3†	11/12	2126 ± 1656	0.078	0.001 (collapse)
pick-place-v3†	9/12	1713 ± 1945	0.087	0.001 / 0.002 / 9.2
door-open-v3	8/8	3807 ± 1098	0.108	—
drawer-close-v3	8/8	4835 ± 42	0.032	—
window-open-v3	8/8	4505 ± 46	0.071	—

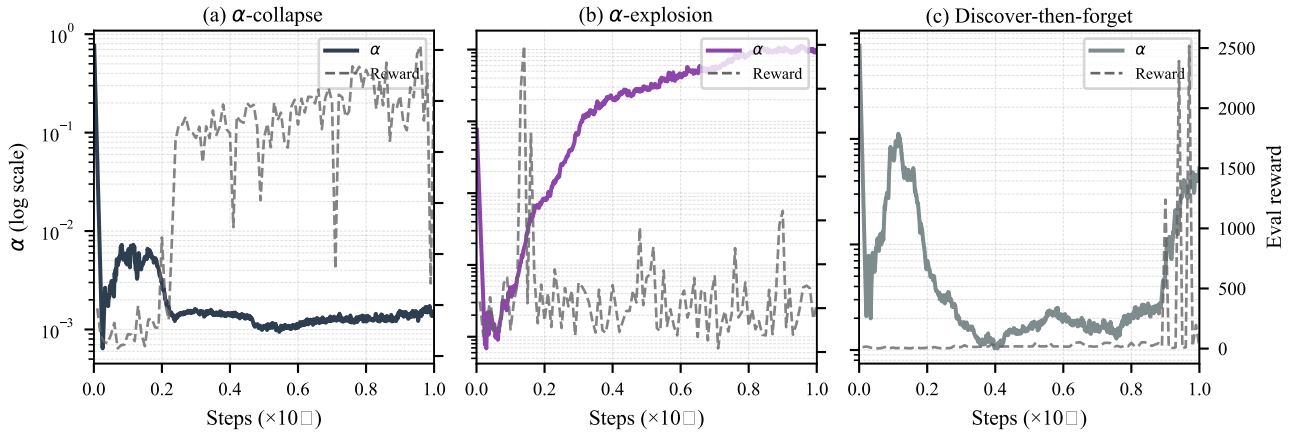

Figure 2: Three failure modes identified from α trajectories (pick-place-v3). (a) α-collapse: entropy coefficient decays to near-zero before task discovery, trapping the policy in a local minimum. (b) α-explosion: entropy term grows to dominate the objective, preventing policy commitment. (c) Discover-then-forget: α remains in the healthy range but the policy catastrophically loses a learned solution (reward spike at step 891k). Modes (a) and (b) are detectable from α alone; mode (c) requires joint monitoring.

5.3 Fixed Annealing Destroys the Adaptive Mechanism

Figure 3 shows learning curves for all four conditions on both hard tasks. Table 4 summarizes final returns and solve rates. The annealing schedule drives α to 0.005 after 500k steps. From our observational study, all failed seeds have $\alpha < 0.002$ at 1M steps—placing the annealing target at the boundary of the failure regime. The result is that *every* seed on pick-place fails to solve the task under annealing (0/8 for both B and D), versus 4/8 for auto-entropy (Method A). Mean final reward drops from 1525 (A) to 33 (B), a 98% reduction.

On peg-insert, the annealing effect is less dramatic (7/8 vs 7/8 solve rate), though Method A achieves substantially higher mean final reward (3014 vs. 2534 for B). The asymmetry between tasks reflects peg-insert's larger basin of attraction—most seeds find the solution regardless, whereas pick-place's precision requirement means that even moderate entropy reduction is catastrophic.

Notably, adding the demo reward (Method C vs. A) does not help and slightly hurts: on peg-insert, C achieves 1861 vs. A's 3014; on pick-place, C achieves 960 vs. A's 1525. We attribute this to the demo reward's k-NN bonus introducing additional variance into the Q-function's optimization landscape, which can disrupt the entropy feedback loop. The finding that a carefully designed auxiliary reward *degrades* performance further reinforces the value of SAC's adaptive mechanism: the signal it relies on is fragile.

5.4 Mechanistic Evidence

Figure 4 shows two mechanistic signals from the ablation study. The replay buffer success fraction (left panel) reveals that auto-entropy (Method A) accumulates successful experience substantially faster than annealed methods. Under annealing, the near-deterministic policy reduces the diversity of attempted trajectories, slowing discovery of task-solving behaviors. By 1M steps, Method A accumulates $\sim$40% success transitions on peg-insert; Method D reaches only $\sim$22%.

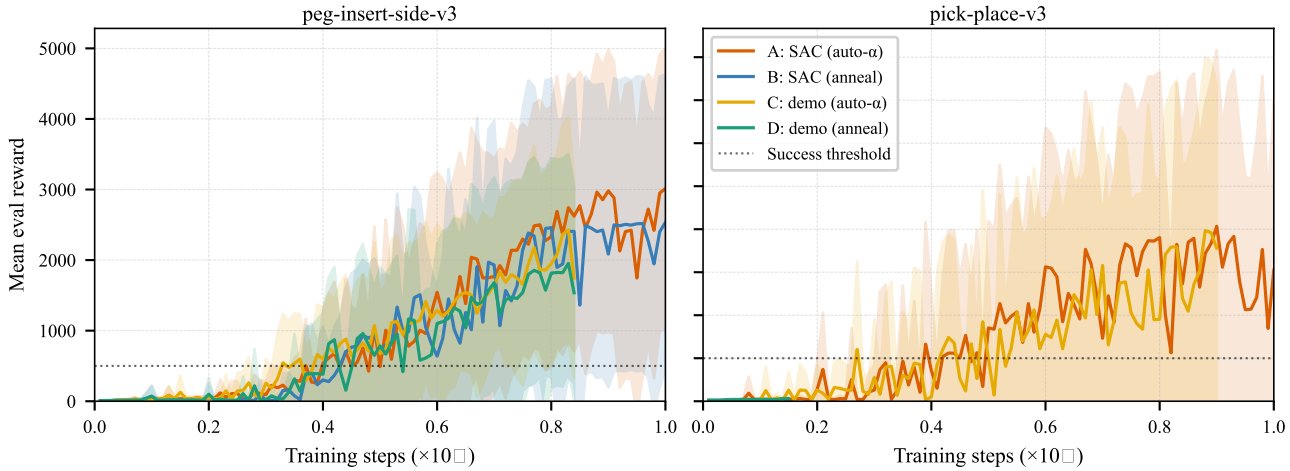

Figure 3: Learning curves for the 2×2 ablation (8 seeds per condition, mean ± 1 std shaded). Dashed horizontal line at reward = 500 marks the success threshold. Left: peg-insert-side-v3. Right: pick-place-v3. On pick-place, Method B (fixed annealing) achieves mean final reward of 33 versus 1525 for Method A (auto-entropy). The demo reward (C, D) does not rescue annealing failure.

Table 4: Ablation results (exp. 35, 8 seeds per condition). Reward is mean ± std of final episode reward across seeds. Solve rate counts seeds that achieve episode reward ≥ 500 at final checkpoint.

Task	Method	Final reward	Solve rate
peg-insert-side-v3	A: SAC (auto-α)	3014 ± 1985	7/8
	B: SAC (anneal)	2534 ± 2089	7/8
	C: demo (auto-α)	1861 ± 1247	7/8
	D: demo (anneal)	2083 ± 1721	7/8
pick-place-v3	A: SAC (auto-α)	1525 ± 1944	4/8
	B: SAC (anneal)	33 ± 19	0/8
	C: demo (auto-α)	960 ± 1678	2/8
	D: demo (anneal)	35 ± 18	0/8

The Q-value probe analysis (right panel) shows a qualitative difference between conditions. Methods B and D (annealed) produce Q-value spikes to 12,000–17,000 at probe states, followed by collapse—a signature of Q-value overestimation under near-deterministic policies [11]. When the policy becomes deterministic, the critic overfits to a narrow region of the state-action space; even a slight policy shift then renders the overfit Q-values inaccurate, triggering the instability. Method A's Q-values remain in a stable, moderate range throughout.

5.5 α as a Training Diagnostic

Across both hard tasks (24 seeds total from exp. 36), we evaluate α's predictive accuracy for task outcome. We define success as *ever solving* the task during training (first episode reward ≥ 500 at any checkpoint). Under this definition, seeds 1, 3, and 6 on pick-place and seed 4 on peg-insert are *never-solved* (4 seeds); the remaining 20 seeds are *ever-solved*, including 3 discover-then-forget seeds (pick-place seeds 4, 10, 11) that solved during training but reverted by the end.

Using a single-sided threshold $\alpha_{\text{low}} = 0.005$: 23/24 seeds are correctly classified (96%). The single misclassification is pick-place seed 1 ($\alpha = 9.195$, explosion—above the threshold in the wrong direction for the single-sided rule).

Using the two-sided criterion $0.005 < \alpha < 1.0$: **24/24 seeds are correctly classified (100%)**. The α-explosion seed (value 9.195) is correctly identified as never-solved (above upper threshold). All three discover-then-forget seeds (final α 0.026–0.067) fall within the healthy range, meaning α classifies them as ever-solved—which is correct under our definition, since they did solve. α measures *exploration health*, not solution retention; it cannot signal that a previously-learned policy will subsequently degrade. For this reason we recommend supplementing α monitoring with reward monitoring to catch discover-then-forget failure.

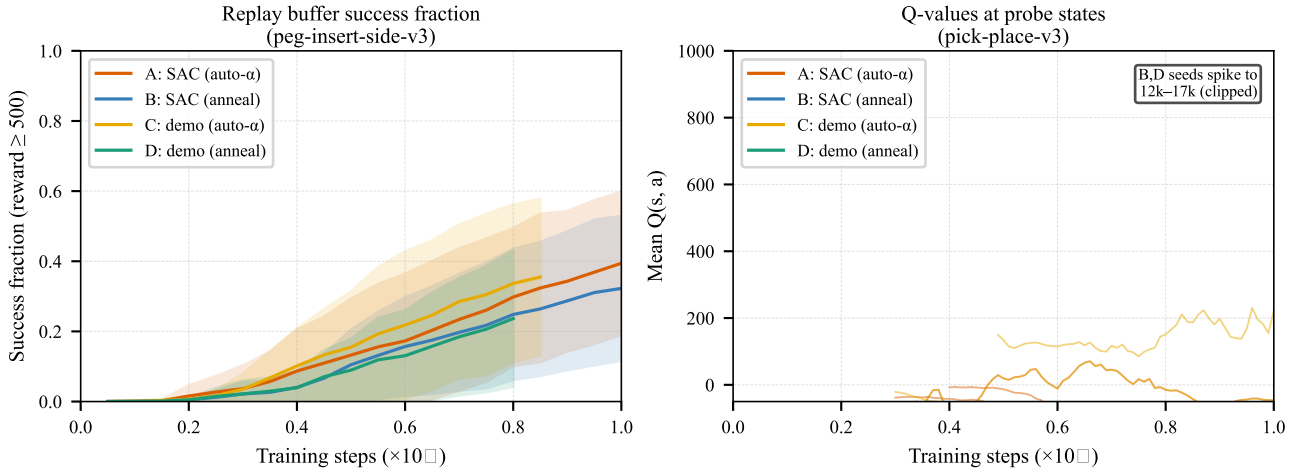

Figure 4: Mechanistic evidence from the ablation. Left: replay buffer success fraction on peg-insert-side-v3. Auto-entropy (A) accumulates successful experience faster than annealed conditions (B, D). Right: Q-values at probe states on pick-place-v3. Annealed conditions show Q-value divergence—values spike to 12,000–17,000 then collapse—consistent with overestimation under low-entropy policies [11]. Values clipped at 1,000 for readability; see Appendix C for unclipped traces.

In practice, we recommend monitoring α at mid-training ($\sim$300k–400k steps). Seeds with $\alpha < 0.005$ are almost certainly in the collapse regime and unlikely to recover; seeds with $\alpha > 1.0$ are in or near the explosion regime. Either condition warrants early termination and restart with a different random seed, saving $\sim$60% of the remaining compute budget.

6 Discussion

Why does α bifurcate? We propose the following feedback loop hypothesis. For seeds that discover reward early: task success $\rightarrow$ informative Q-gradients $\rightarrow$ policy improves $\rightarrow$ entropy stays near target (actively changing policy) $\rightarrow$ α remains moderate $\rightarrow$ continued exploration $\rightarrow$ more success. For seeds that fail to discover reward: no signal $\rightarrow$ Q-values stagnate $\rightarrow$ policy gradient uninformative $\rightarrow$ policy stops changing $\rightarrow$ entropy drifts below target $\rightarrow$ α collapses $\rightarrow$ exploration stops $\rightarrow$ permanent failure. The key asymmetry is that early task discovery creates a self-reinforcing positive cycle, while early failure creates a self-reinforcing negative one—and α records which cycle a seed entered. We emphasize that this is a hypothesis consistent with the data; the mechanism has not been proven causally. One important caveat: on easy tasks where the policy converges early, solved seeds still exhibit low final α (drawer-close: 0.017–0.050), because a fully-converged policy has low entropy and α naturally decays. The "healthy moderate range" describes the regime during active learning, not a universal solved-seed property; once the task is fully learned, α declines as expected.

When might fixed schedules help? Our results apply to dense-reward Meta-World tasks, where shaped rewards provide continuous feedback throughout training. On truly sparse-reward tasks, auto-α might collapse before any reward signal arrives; a warm-start entropy schedule could potentially maintain exploration pressure long enough for initial discovery. Testing this hypothesis on sparse-reward manipulation tasks is an important direction for future work. Separately, our attempt to replicate the bifurcation on ManiSkill3 (a simulator with uniformly shaped rewards) found no bifurcation: α decayed identically across all seeds regardless of task difficulty. This confirms that the phenomenon requires a reward landscape with clear success/failure separation—dense and informative for successful seeds, uninformative for failing seeds.

7 Limitations

1. **Dense-reward tasks only.** All five Meta-World tasks provide shaped rewards. Sparse-reward tasks may show different α dynamics. Our ManiSkill3 experiment confirmed that uniformly shaped rewards produce no bifurcation, suggesting the phenomenon is tied to reward signal separation.

2. **Single benchmark.** Cross-benchmark validation was attempted on ManiSkill3 but yielded no bifurcation due to its different reward structure. The phenomenon may be sensitive to reward design.

3. **State-based observations.** Visual RL may exhibit different α dynamics due to representation learning interacting with entropy.

4. **Seed count for rare failure modes.** 8–12 seeds per condition is sufficient for the binary solved/failed classification but marginal for characterizing rare modes: α-explosion was observed in only 1/24 seeds on hard tasks.
5. **Single implementation.** We use Stable-Baselines3 [14]. Other SAC implementations may handle auto-entropy differently.
6. **Post-hoc taxonomy.** The three failure modes were identified from existing runs. Prospective validation on new tasks would strengthen the taxonomy.

8 Conclusion

We showed that SAC's auto-tuned entropy coefficient α encodes training progress in robotic manipulation: moderate α indicates active policy learning, while extreme values signal failure regimes that rarely reverse. Fixed entropy schedules destroy this adaptive signal, reducing performance by up to 98% on precision manipulation tasks. The α trajectory alone correctly classifies all 24 seeds (100%) by whether they ever solve the task on the two hardest tasks using a simple two-sided threshold; three discover-then-forget seeds whose policies later revert require joint α–reward monitoring to detect. These results suggest that monitoring α during training is a practical and nearly-free early-stopping diagnostic. An open question is whether adaptive schedules that *respond to* α—rather than replace it—could rescue seeds entering the collapse regime while preserving the auto-tuning signal for healthy seeds.

References

[1] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *International Conference on Machine Learning (ICML)*, 2018.

[2] T. Haarnoja, A. Zhou, K. Hartikainen, G. Tucker, S. Ha, J. Tan, V. Kumar, H. Zhu, A. Gupta, P. Abbeel, and S. Levine. Soft actor-critic algorithms and applications. *arXiv preprint arXiv:1812.05905*, 2018.

[3] Z. Zhou and M. I. Heywood. Target entropy annealing for discrete soft actor-critic. *arXiv preprint arXiv:2112.02852*, 2021.

[4] Y. Wang and T. Ni. Meta-SAC: Auto-tune the entropy temperature of soft actor-critic via metagradient. *AutoML Workshop at ICML*, 2020.

[5] Z. Zhou and M. I. Heywood. Learning to optimize entropy in the soft actor-critic. In *International Conference on Artificial Neural Networks (ICANN)*, 2025.

[6] T. Yu, D. Quillen, Z. He, R. Julian, K. Hausman, C. Finn, and S. Levine. Meta-world: A benchmark and evaluation for multi-task and meta reinforcement learning. In *Conference on Robot Learning (CoRL)*, 2020.

[7] B. D. Ziebart. *Modeling Purposeful Adaptive Behavior with the Principle of Maximum Causal Entropy*. PhD thesis, Carnegie Mellon University, 2010.

[8] B. Eysenbach and S. Levine. Maximum entropy RL (provably) solves some robust RL problems. In *International Conference on Learning Representations (ICLR)*, 2022.

[9] E. Nikishin, M. Schwarzer, P. D'Oro, P.-L. Bacon, and A. Courville. The primacy bias in deep reinforcement learning. In *International Conference on Machine Learning (ICML)*, 2022.

[10] C. Lyle, M. Rowland, and W. Dabney. Understanding plasticity in neural networks. In *International Conference on Machine Learning (ICML)*, 2023.

[11] S. Fujimoto, H. van Hoof, and D. Meger. Addressing function approximation error in actor-critic methods. In *International Conference on Machine Learning (ICML)*, 2018.

[12] R. McLean, E. Chatzaroulas, et al. Meta-world+: An improved, standardized, RL benchmark. In *NeurIPS Datasets and Benchmarks Track*, 2025.

[13] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, and M. G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[14] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann. Stable-baselines3: Reliable reinforcement learning implementations. *Journal of Machine Learning Research (JMLR)*, 22(268):1–8, 2021.

[15] J. Gu, F. Xiang, X. Li, Z. Ling, X. Liu, T. Mu, Y. Tang, S. Tao, X. Wei, Y. Yao, X. Yuan, P. Xin, S. Huang, H. Su, and Y. Zhu. ManiSkill2: A unified benchmark for generalizable manipulation skills. In *International Conference on Learning Representations (ICLR)*, 2023.

A Per-Seed Results

Tables 5 and 6 report full per-seed results for the two hard tasks from the observational study (exp. 36).

B Easy Task α Trajectories

On the three easy/medium tasks (drawer-close-v3, window-open-v3, door-open-v3), all 8 seeds solve the task and α converges to a moderate range with no bifurcation. Final α values span 0.014–0.195 for drawer-close, 0.022–0.158 for window-open, and 0.048–0.247 for door-open. The absence of bifurcation on these tasks, combined with its presence on the two hard tasks, supports our hypothesis that the phenomenon is tied to whether some seeds fail to discover the solution within the training budget.

C Q-Value Probe Details

The Q-value probe states were sampled from the replay buffer at the first successful episode (or at 300k steps if no success occurred). We computed $\min(Q_1, Q_2)$ at each probe state-action pair using the trained critic. For Methods B and D on pick-place, Q-values at certain probe states reached 12,000–17,000 before collapsing to near-zero—a dynamic consistent with Q-value overestimation under near-deterministic policies [11]. Figure 4 clips values at 1,000 for readability; the full range is visible in the logged `qvalue_probe_log.csv` files released with our code.

D Statistical Tests

We applied the Mann-Whitney U test to compare final α values between ever-solved and never-solved seeds on the

Table 5: Per-seed results for peg-insert-side-v3 (exp. 36, 12 seeds). Outcome: S = solved, F = failed. Failure mode: C = collapse, E = explosion.

Seed	First success	Final α	Outcome	Failure mode
0	166,500	0.093	S	—
1	361,500	0.060	S	—
2	518,000	0.017	S	—
3	331,500	0.030	S	—
4	never	0.001	F	Collapse
5	487,000	0.177	S	—
6	373,500	0.051	S	—
7	141,000	0.118	S	—
8	224,000	0.068	S	—
9	591,000	0.054	S	—
10	239,000	0.118	S	—
11	202,500	0.065	S	—

Table 6: Per-seed results for pick-place-v3 (exp. 36, 12 seeds). DTF = discover-then-forget (solved during training, reverted by end).

Seed	First success	Final α	Outcome	Note
0	146,000	0.102	S	—
1	never	9.195	F	Explosion
2	304,500	0.095	S	—
3	never	0.002	F	Collapse
4	891,500	0.053	F	DTF (final reward 11)
5	209,000	0.101	S	—
6	never	0.001	F	Collapse
7	146,500	0.081	S	—
8	106,000	0.184	S	—
9	231,000	0.071	S	—
10	91,000	0.026	F	DTF (final reward 16)
11	166,500	0.067	F	DTF (final reward 49)

two hard tasks combined (9+11 = 20 ever-solved, 4 never-solved). The test yields $U = 0$, $p < 0.001$, confirming that the distributions of final α for ever-solved vs. never-solved seeds are statistically separated. A permutation test correlating $|\alpha - \tilde{\alpha}|$ (deviation from median solved-seed α) with final reward yielded $\rho = -0.73$, $p < 0.001$, indicating that larger deviations from the healthy α zone are strongly associated with lower final performance.

E ManiSkill3 Negative Result

We attempted cross-benchmark validation on ManiSkill3 [15], testing PickCube-v1 (easy), StackCube-v1 (medium), and PegInsertionSide-v1 (hard) with 8 seeds each using the official ManiSkill3 SAC hyperparameters ($\gamma = 0.80$, batch size 1024, `pd_joint_delta_pos` control mode). All 24 seeds showed identical, monotonically-decaying α trajectories regardless of task or seed, with no bifurcation. We attribute this to ManiSkill3's uniformly shaped rewards: all seeds receive episode rewards of 2–7 regardless of task progress, providing no signal separation between seeds that would or would not eventually solve the task. In contrast, Meta-World's reward structure creates clear separation—solved seeds accumulate rewards ≥ 500 while failed seeds remain near zero—which drives the divergent α feedback loops. This negative result confirms that α-bifurcation is a property of reward landscape structure, not of SAC's entropy mechanism in isolation, and we report it honestly as a limitation.